

1. How does Shipline's expected share price change under the analyst's constant growth assumption?

Solution:

GBP17.67

Using Equation 14, solve for PV using D_{t+1} of GBP1.59 ($=\text{GBP}1.50 \times (1 + 0.06)$), r of 15%, and g of 6%:

$$PV = \text{GBP}1.59 / (15\% - 6\%) = \text{GBP}17.67.$$

Note that a higher growth rate g increases the PV by reducing the denominator ($r - g$).

2. How does Shipline's expected share price change if we assume instead an initial dividend growth of 6 percent over a three-year period followed by constant 2 percent dividend growth thereafter?

Solution:

GBP13.05

We may solve for Shipline's expected share price (PV) using Equations 15–17 and $D_t = 1.5$, $g_s = 6\%$, $n = 3$, $r = 15\%$, and $g_l = 2\%$, as follows:

$$PV_t = \sum_{i=1}^3 \frac{1.50(1 + 0.06)^i}{(1 + 0.15)^i} + \sum_{j=4}^{\infty} \frac{D_{t+3}(1 + 0.02)^j}{(1 + 0.15)^j}.$$

As a first step, we calculate the present value of dividends associated with the higher 6 percent growth rate over the first three years as shown in the first expression. As a second step, we calculate the present value of future dividends at a lower 2 percent growth rate for an indefinite period. The sum of these two steps is the expected share price.

- Step 1 Solve for the first step as GBP3.832 with dividend cash flows for the initial growth period as shown in the following spreadsheet:

t	D	$PV(D)$ at $r\%$
0	1.500	
1	1.590	1.383
2	1.685	1.274
3	1.787	1.175
$PV(ST \text{ growth})$		3.832

- Step 2 As shown in Equation 17, the second expression simplifies as follows:

$$\frac{E(S_4)}{(1 + r)^3}; \text{ with } E(S_4) = \frac{D_4}{r - g_l}.$$

- We may solve for D_4 as GBP1.894 ($=1.787 \times 1.02 = D_3(1 + g_l)$) and the second expression to be GBP9.22 as follows:

$$\text{GBP}9.22 = \frac{1.894(0.15 - 0.02)}{(1.15)^3}.$$

- Step 3 The sum of these two steps gives us an expected Shipline share price of GBP13.05 ($=3.83 + 9.22$).

In the following table, we show the sensitivity of Shipline's expected share price (PV) to changes in the long-term growth rate (g_l) after three years of 6 percent dividend growth: